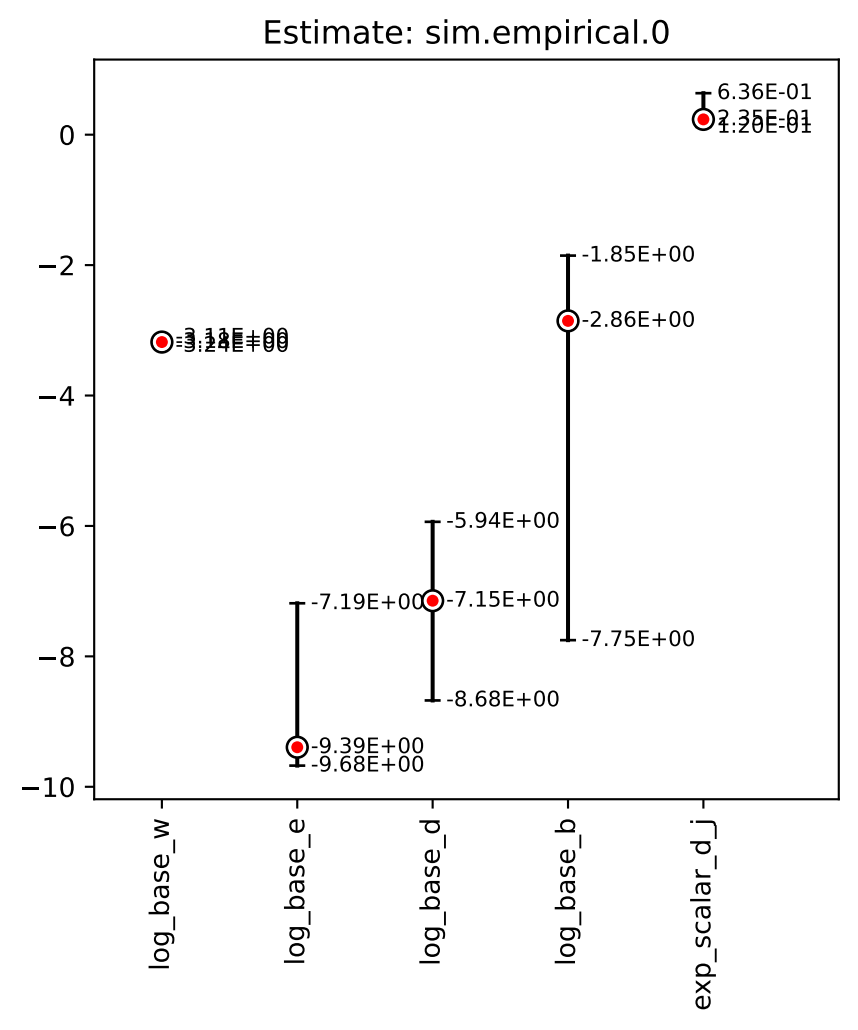
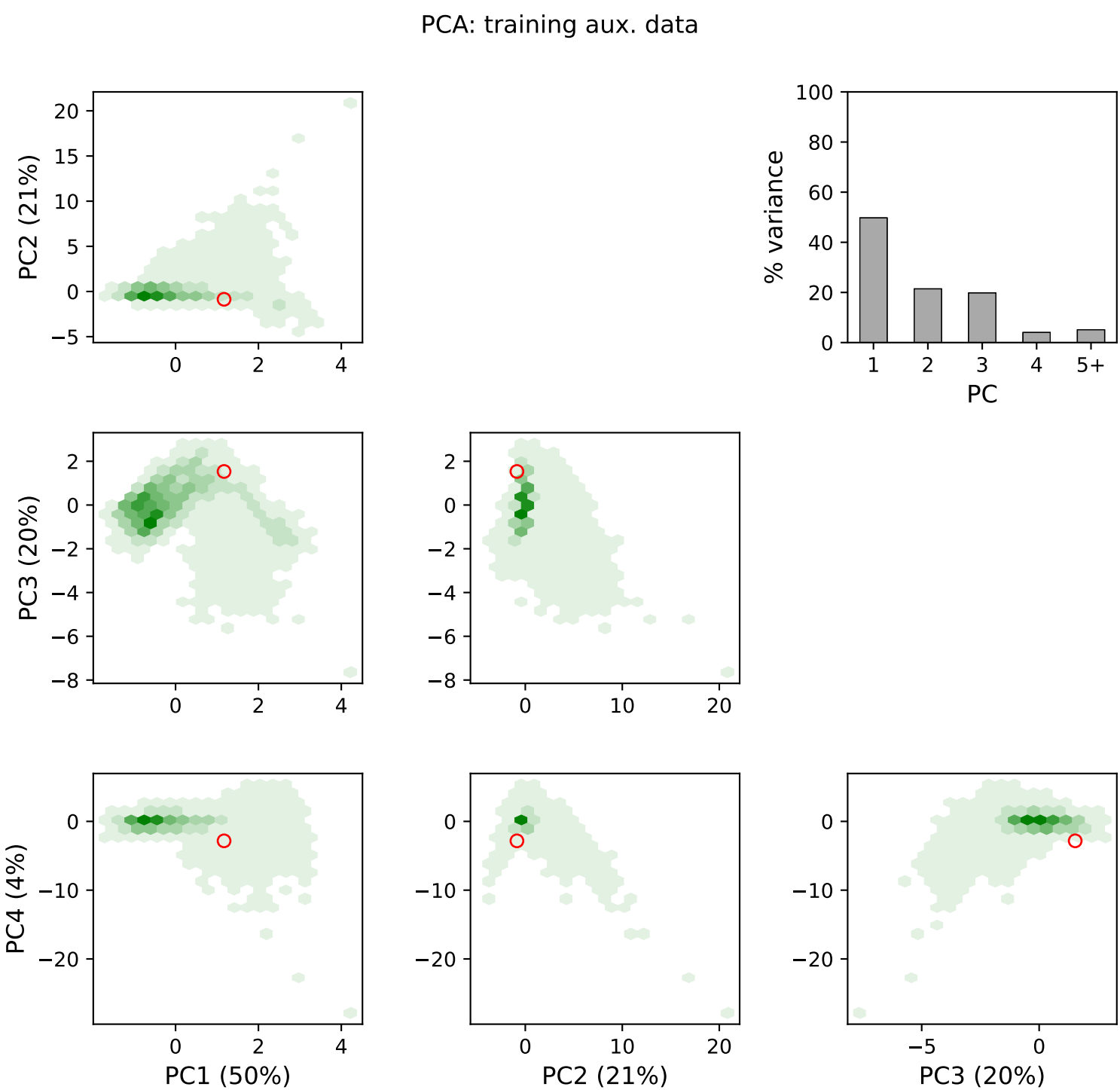
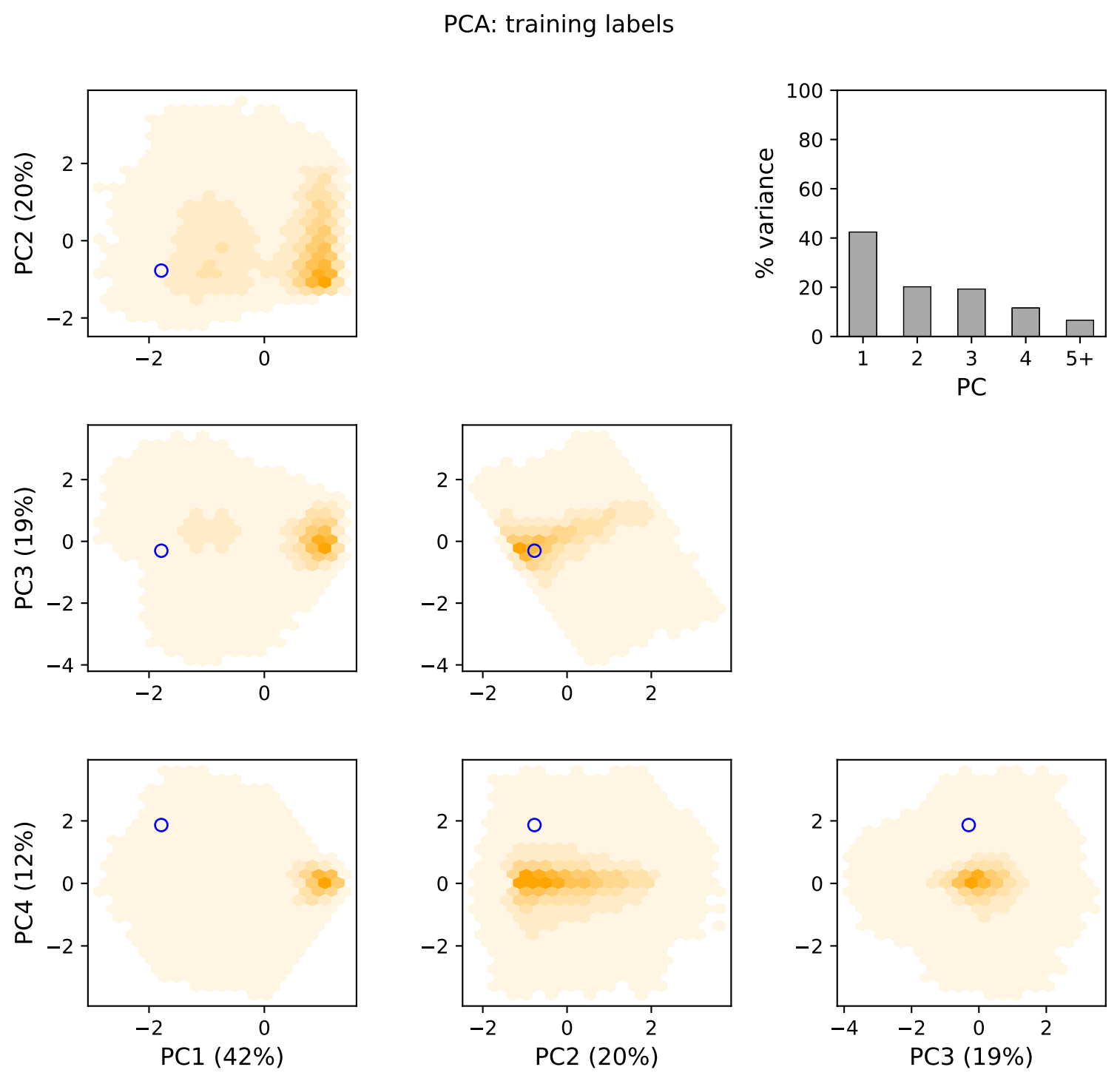
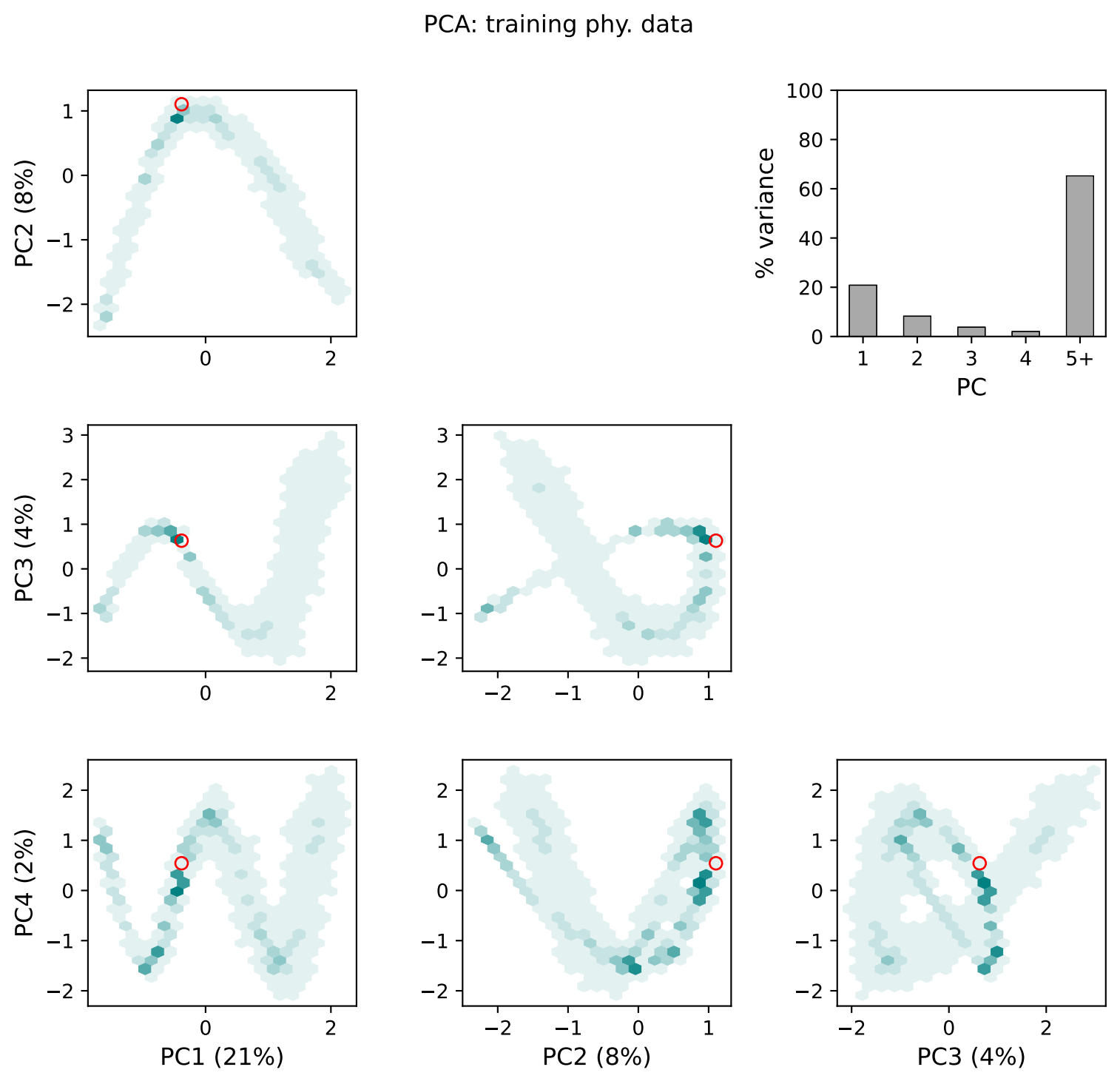
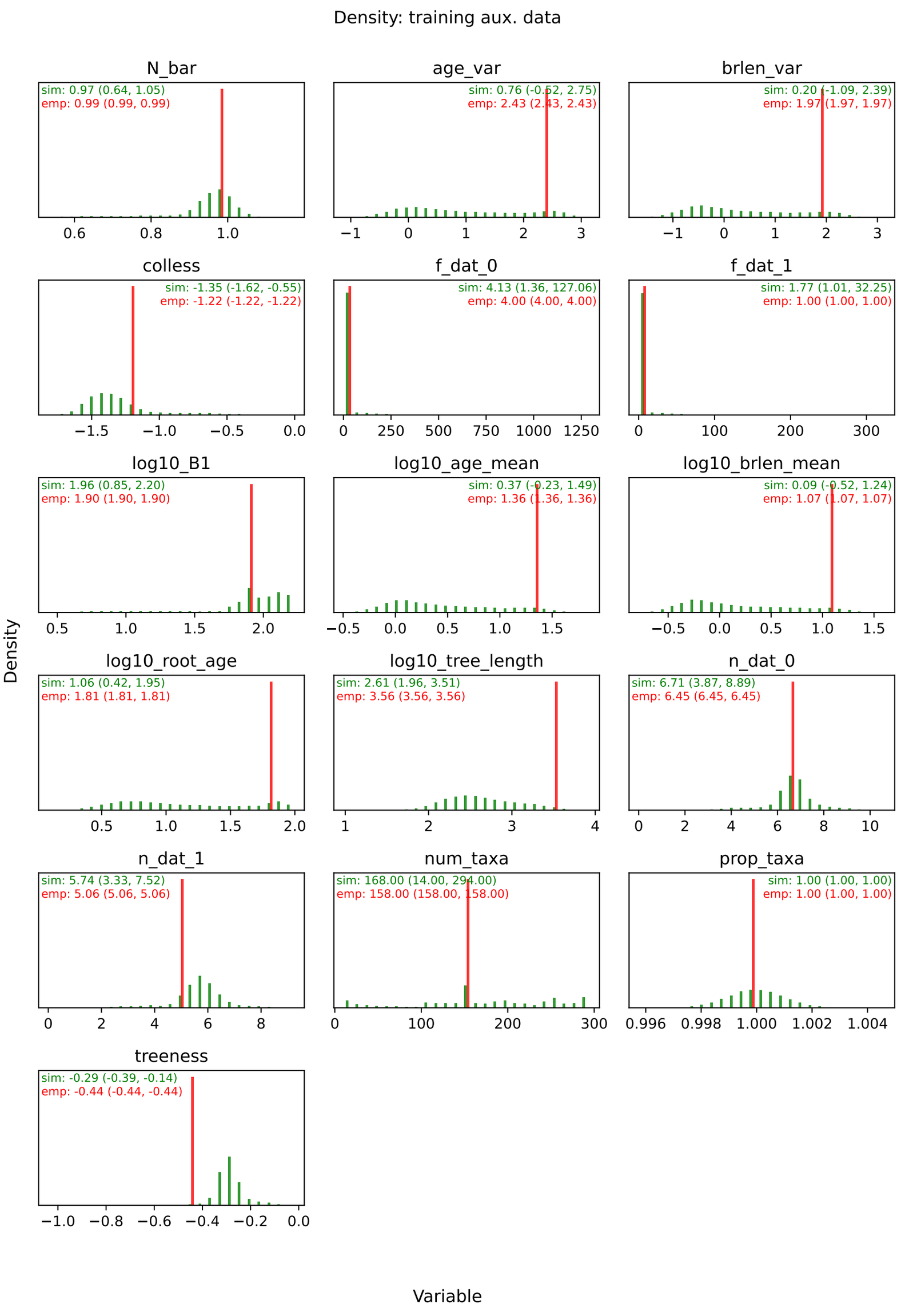
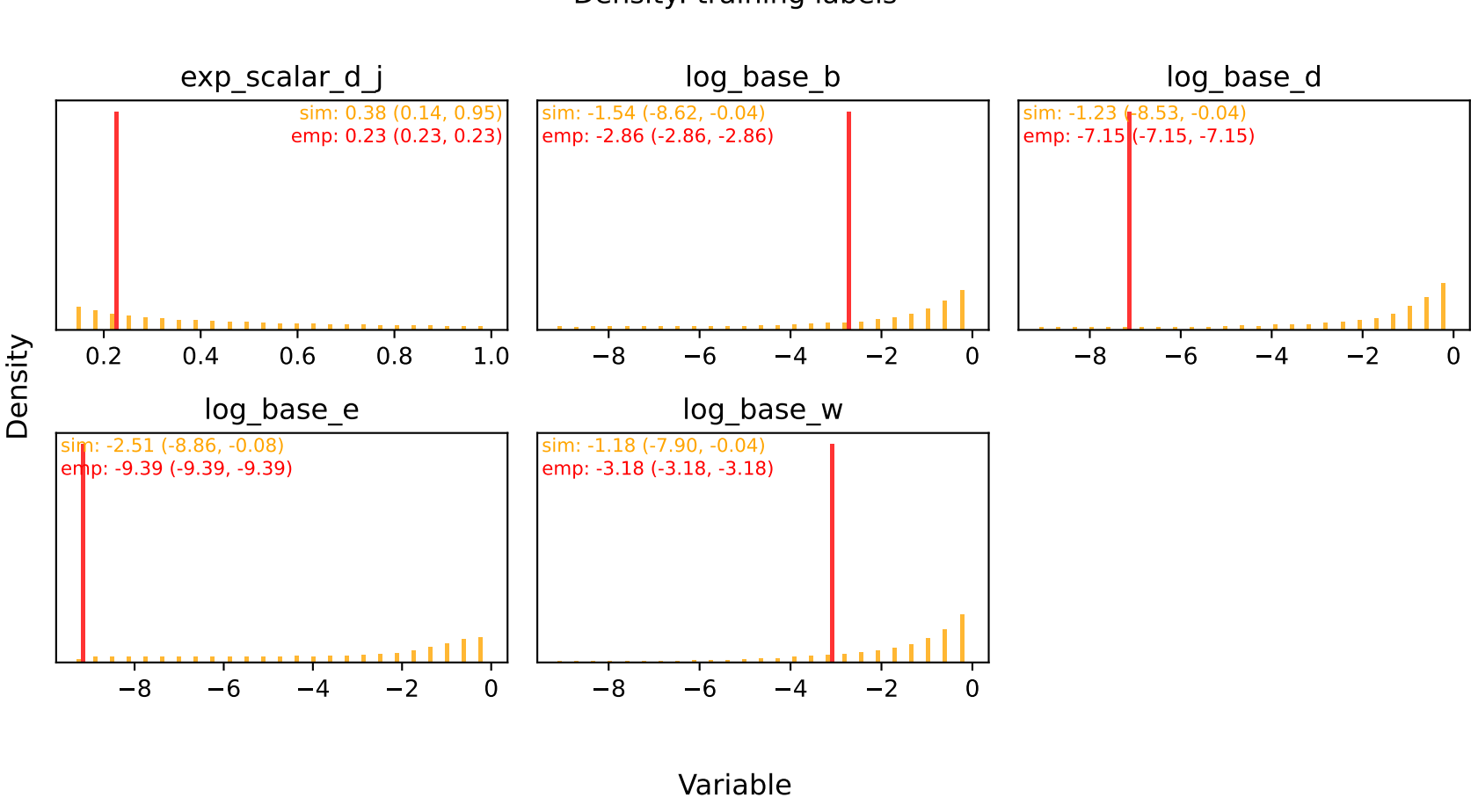

Estimate: sim.empirical.0
6.36E-01
2.35E-01
1.20E-01
-1.85E+00
-2.86E+00
-7.75E+00
-5.94E+00
-7.19E+00
-7.15E+00
-8.68E+00
-9.39E+00
-9.68E+00
0
−2
−4
−6
−8
−10
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_d_j

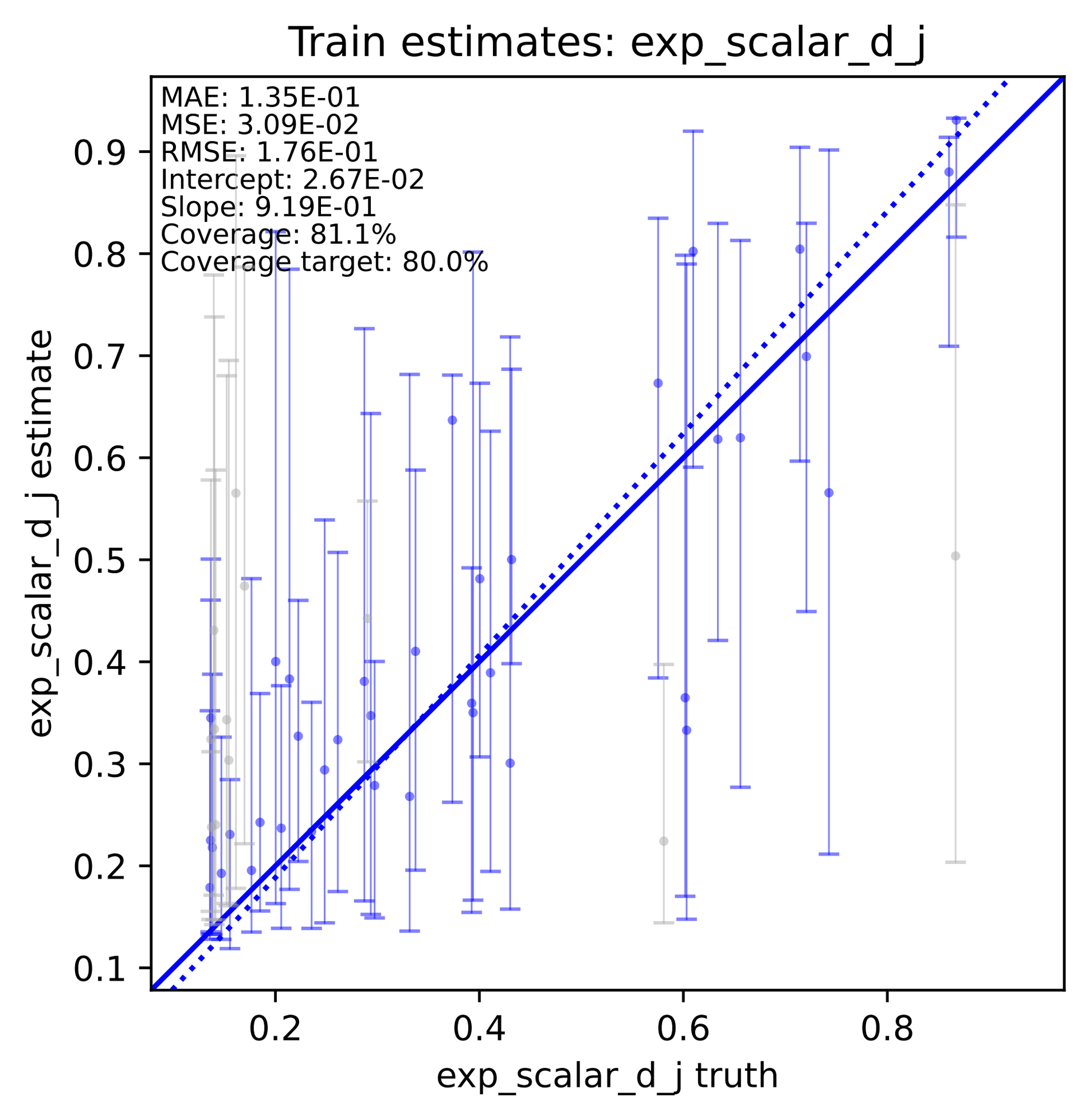

PCA: training aux. data
PC2 (21%)
% variance
PC
PC3 (20%)
PC4 (4%)
PC1 (50%)
PC2 (21%)
PC3 (20%)

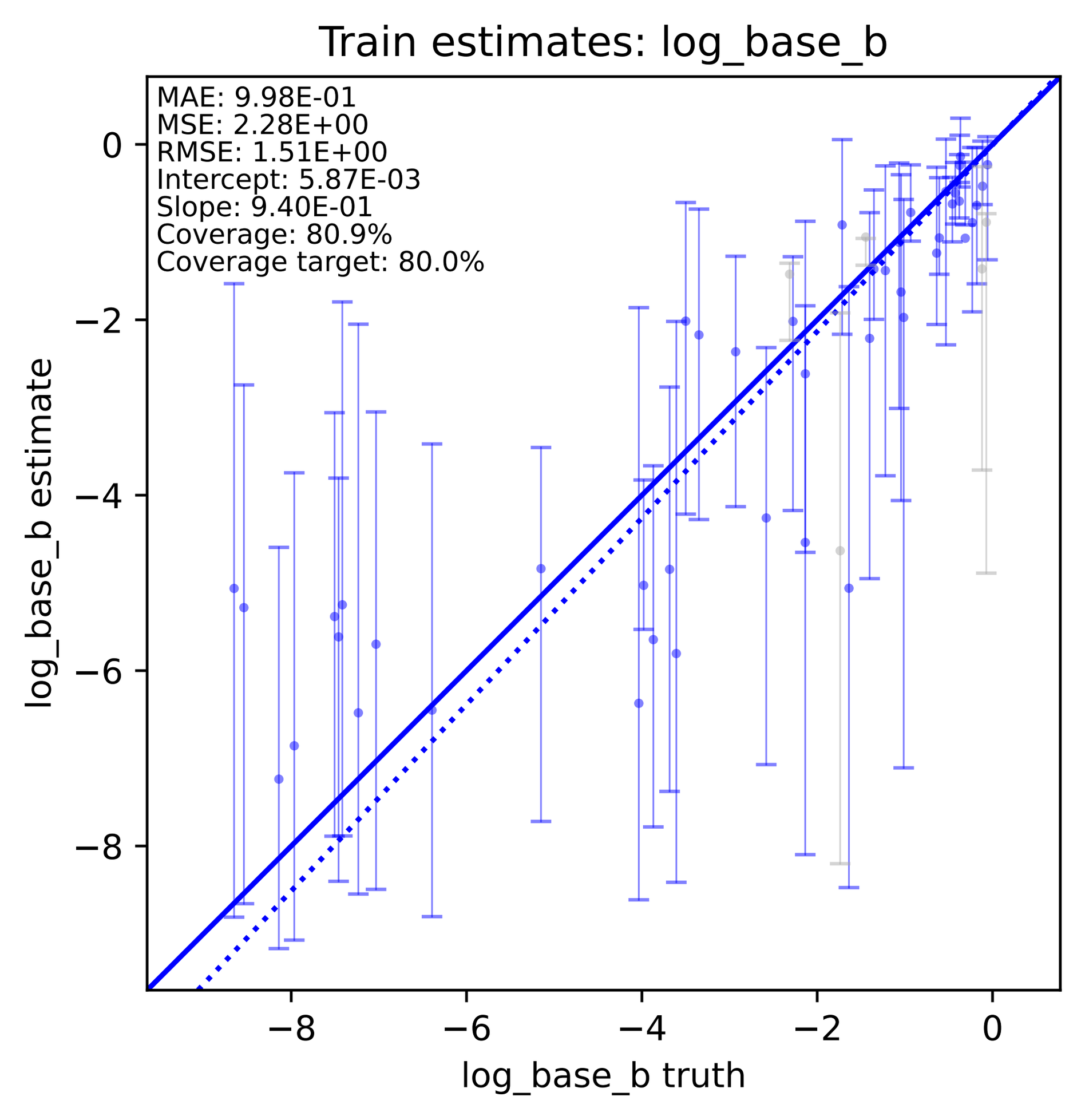

PCA: training labels
PC2 (20%)
% variance
PC
PC3 (19%)
PC4 (12%)
PC1 (42%)
PC2 (20%)
PC3 (19%)

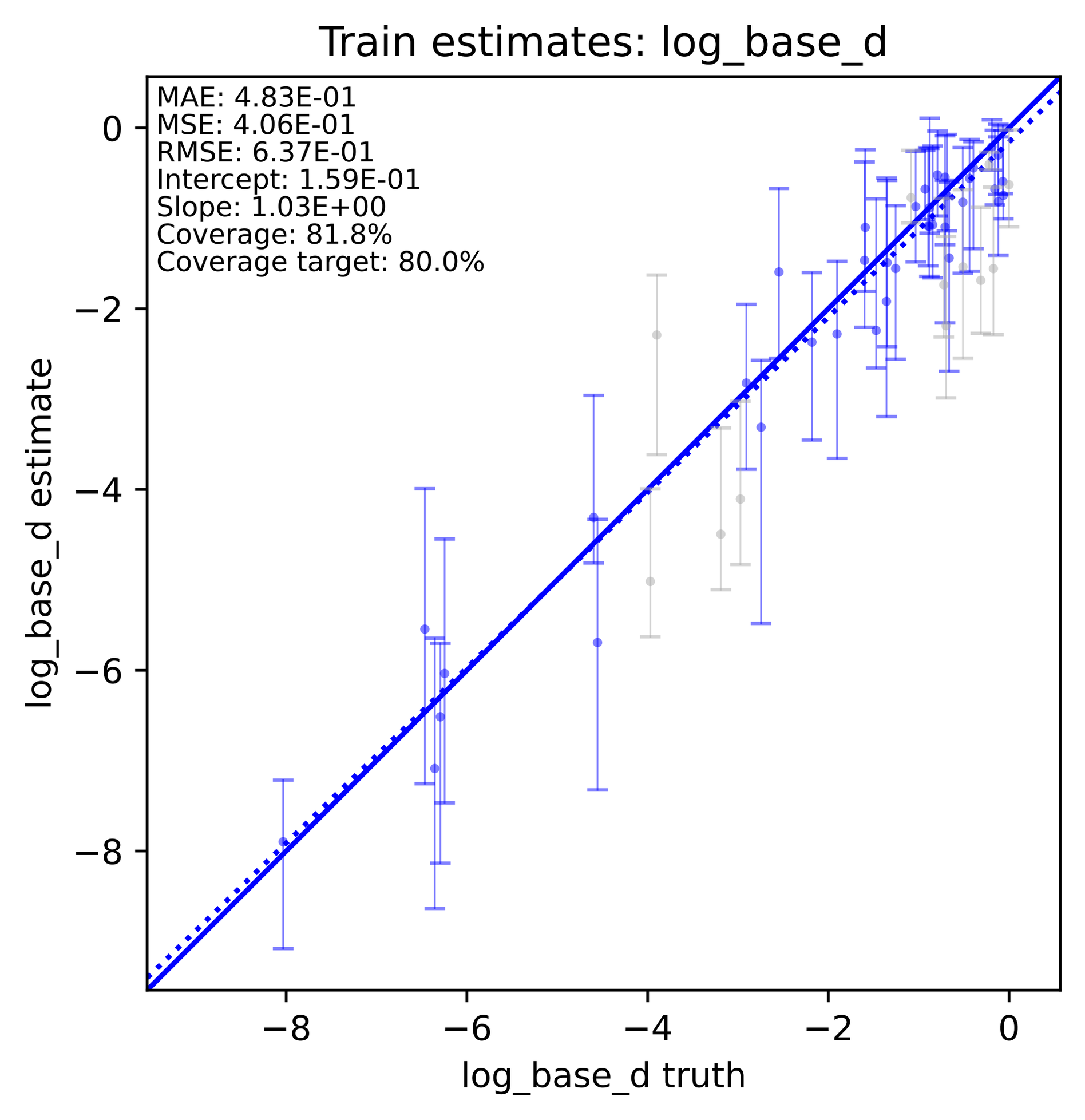

PCA: training phy. data
PC2 (8%)
% variance
PC
PC3 (4%)
PC4 (2%)
PC1 (21%)
PC2 (8%)
PC3 (4%)

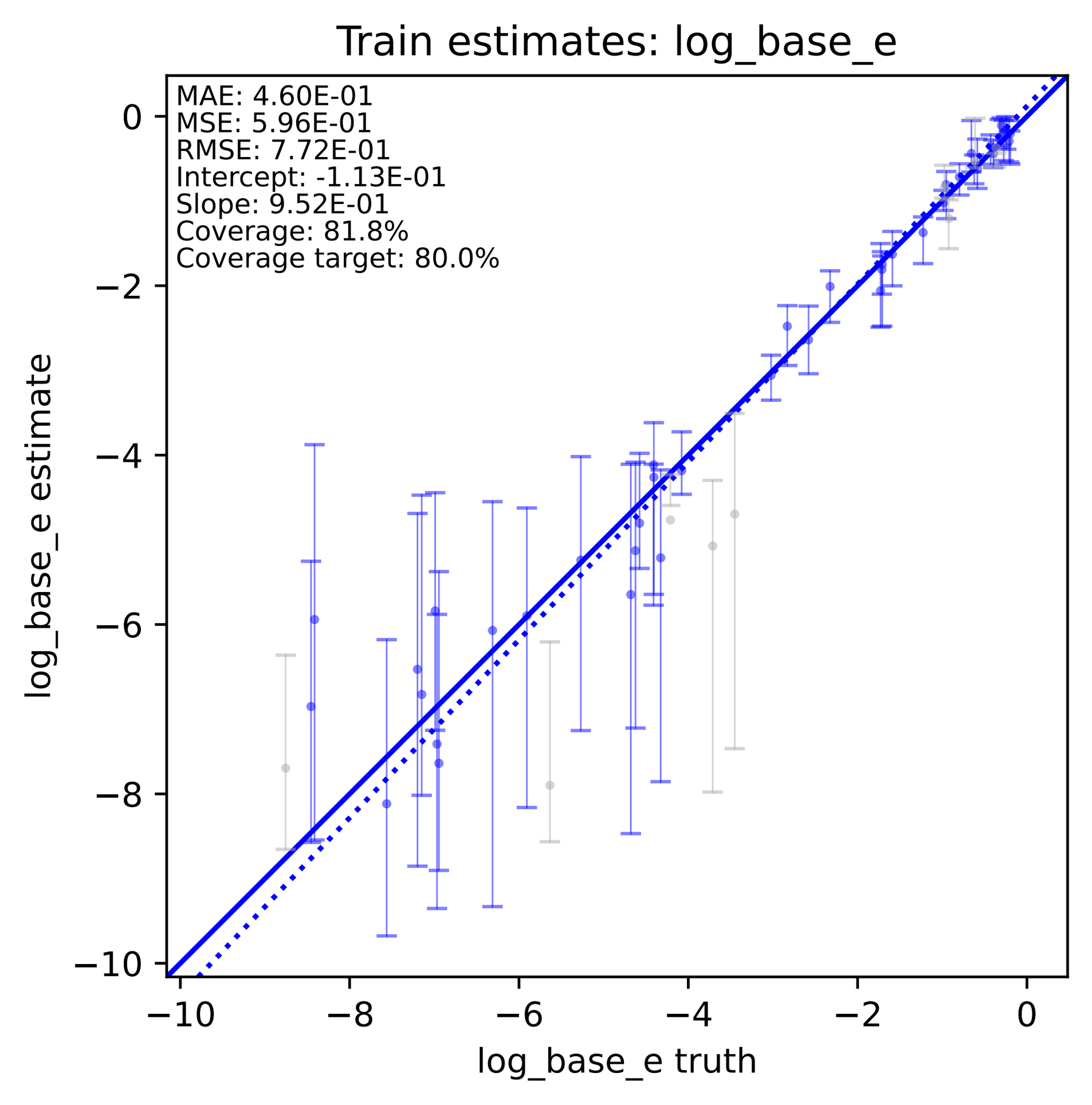

Density: training aux. data
N_bar
sim: 0.97 (0.64, 1.05)
emp: 0.99 (0.99, 0.99)
age_var
sim: 0.76 (-0.52, 2.75)
emp: 2.43 (2.43, 2.43)
brlen_var
sim: 0.20 (-1.09, 2.39)
emp: 1.97 (1.97, 1.97)
colless
sim: -1.35 (-1.62, -0.55)
emp: -1.22 (-1.22, -1.22)
f_dat_0
sim: 4.13 (1.36, 127.06)
emp: 4.00 (4.00, 4.00)
f_dat_1
sim: 1.77 (1.01, 32.25)
emp: 1.00 (1.00, 1.00)
log10_B1
sim: 1.96 (0.85, 2.20)
emp: 1.90 (1.90, 1.90)
log10_age_mean
sim: 0.37 (-0.23, 1.49)
emp: 1.36 (1.36, 1.36)
log10_brlen_mean
sim: 0.09 (-0.52, 1.24)
emp: 1.07 (1.07, 1.07)
log10_root_age
sim: 1.06 (0.42, 1.95)
emp: 1.81 (1.81, 1.81)
log10_tree_length
sim: 2.61 (1.96, 3.51)
emp: 3.56 (3.56, 3.56)
n_dat_0
sim: 6.71 (3.87, 8.89)
emp: 6.45 (6.45, 6.45)
n_dat_1
sim: 5.74 (3.33, 7.52)
emp: 5.06 (5.06, 5.06)
num_taxa
sim: 168.00 (14.00, 294.00)
emp: 158.00 (158.00, 158.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.29 (-0.39, -0.14)
emp: -0.44 (-0.44, -0.44)
Density
Variable

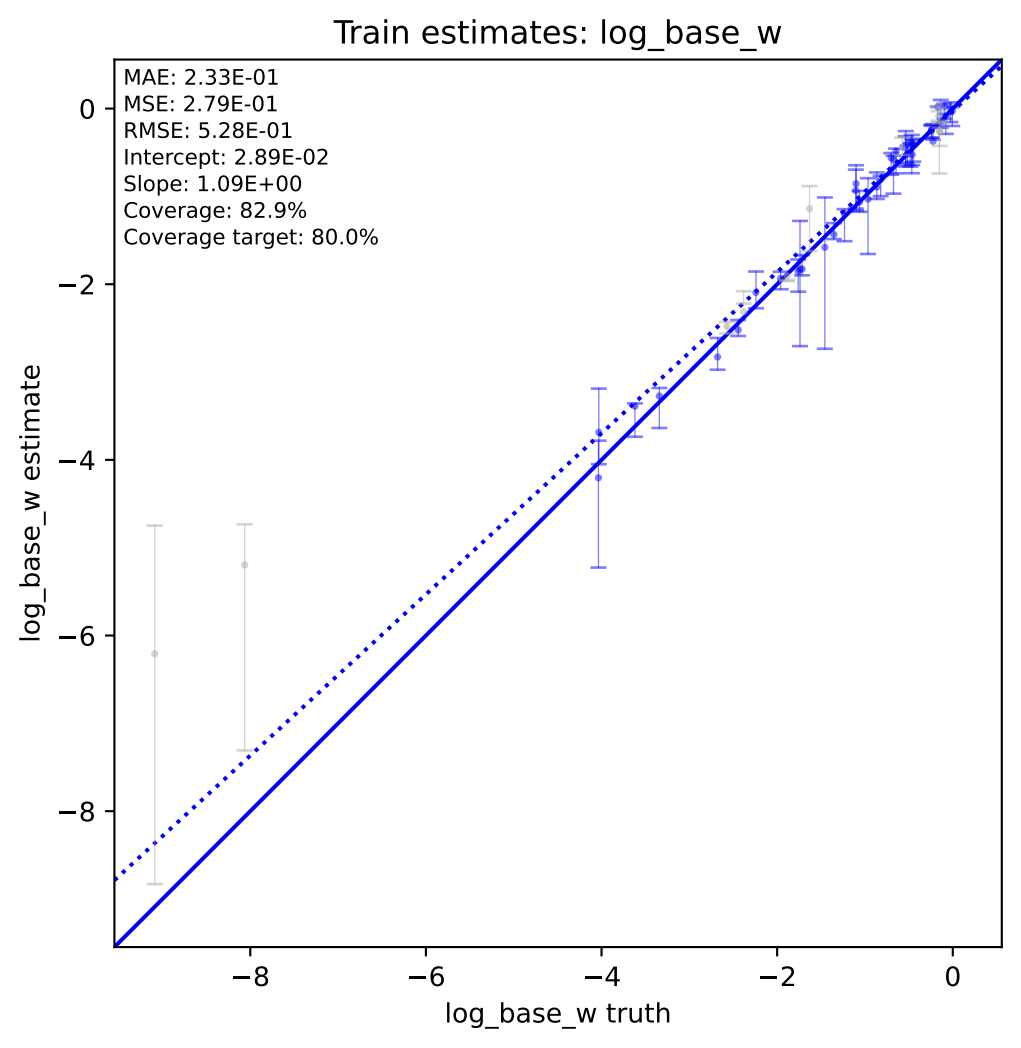

Density: training labels
exp_scalar_d_j
sim: 0.38 (0.14, 0.95)
emp: 0.23 (0.23, 0.23)
log_base_b
sim: -1.54 (-8.62, -0.04)
emp: -2.86 (-2.86, -2.86)
log_base_d
sim: -1.23 (-8.53, -0.04)
emp: -7.15 (-7.15, -7.15)
log_base_e
sim: -2.51 (-8.86, -0.08)
emp: -9.39 (-9.39, -9.39)
log_base_w
sim: -1.18 (-7.90, -0.04)
emp: -3.18 (-3.18, -3.18)
Density
Variable

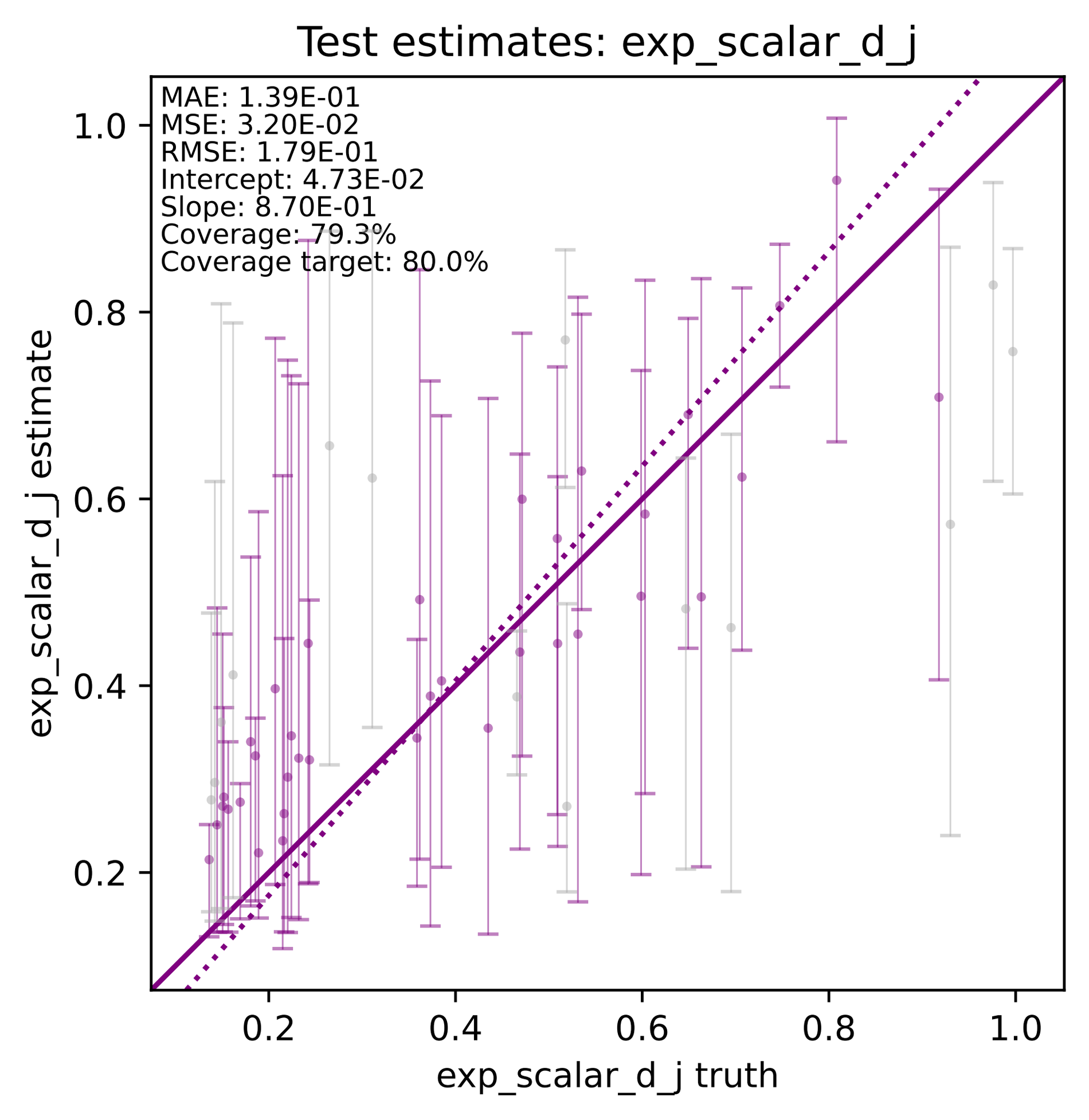

Train estimates: exp_scalar_d_j
MAE: 1.35E-01
MSE: 3.09E-02
RMSE: 1.76E-01
Intercept: 2.67E-02
Slope: 9.19E-01
Coverage: 81.1%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
0.2
0.4
0.6
0.8

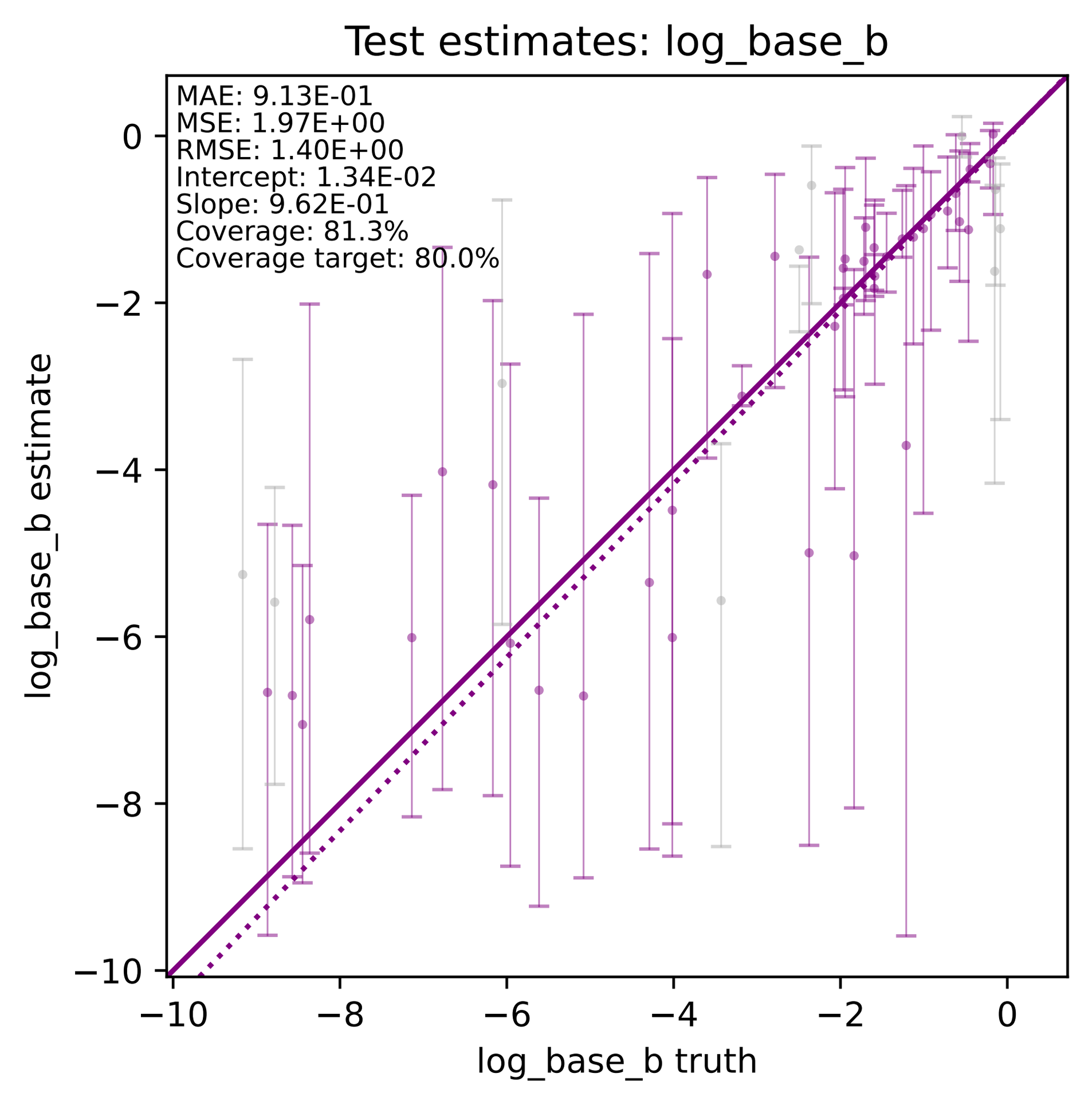

Train estimates: log_base_b
MAE: 9.98E-01
MSE: 2.28E+00
RMSE: 1.51E+00
Intercept: 5.87E-03
Slope: 9.40E-01
Coverage: 80.9%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

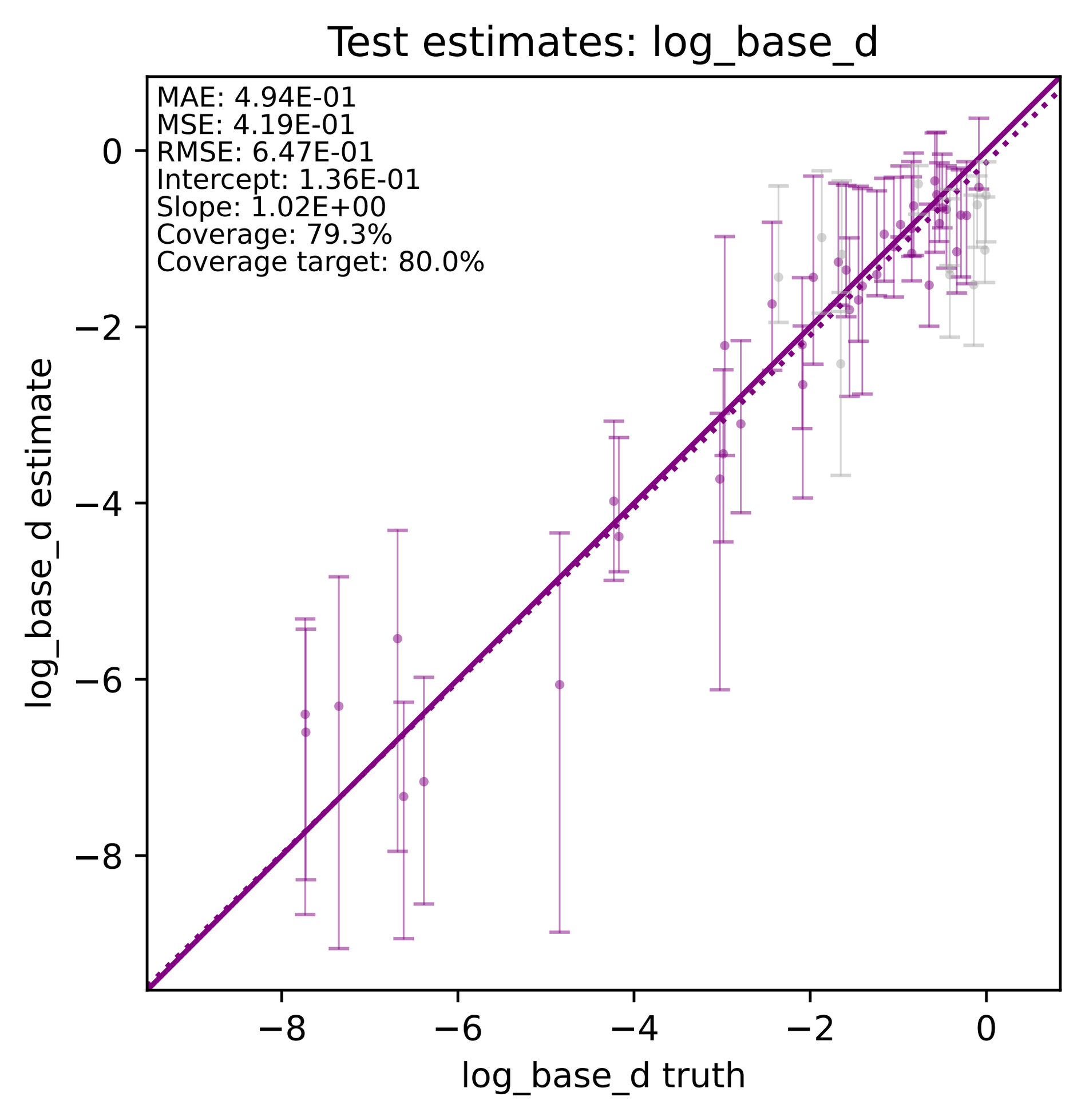

Train estimates: log_base_d
MAE: 4.83E-01
MSE: 4.06E-01
RMSE: 6.37E-01
Intercept: 1.59E-01
Slope: 1.03E+00
Coverage: 81.8%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

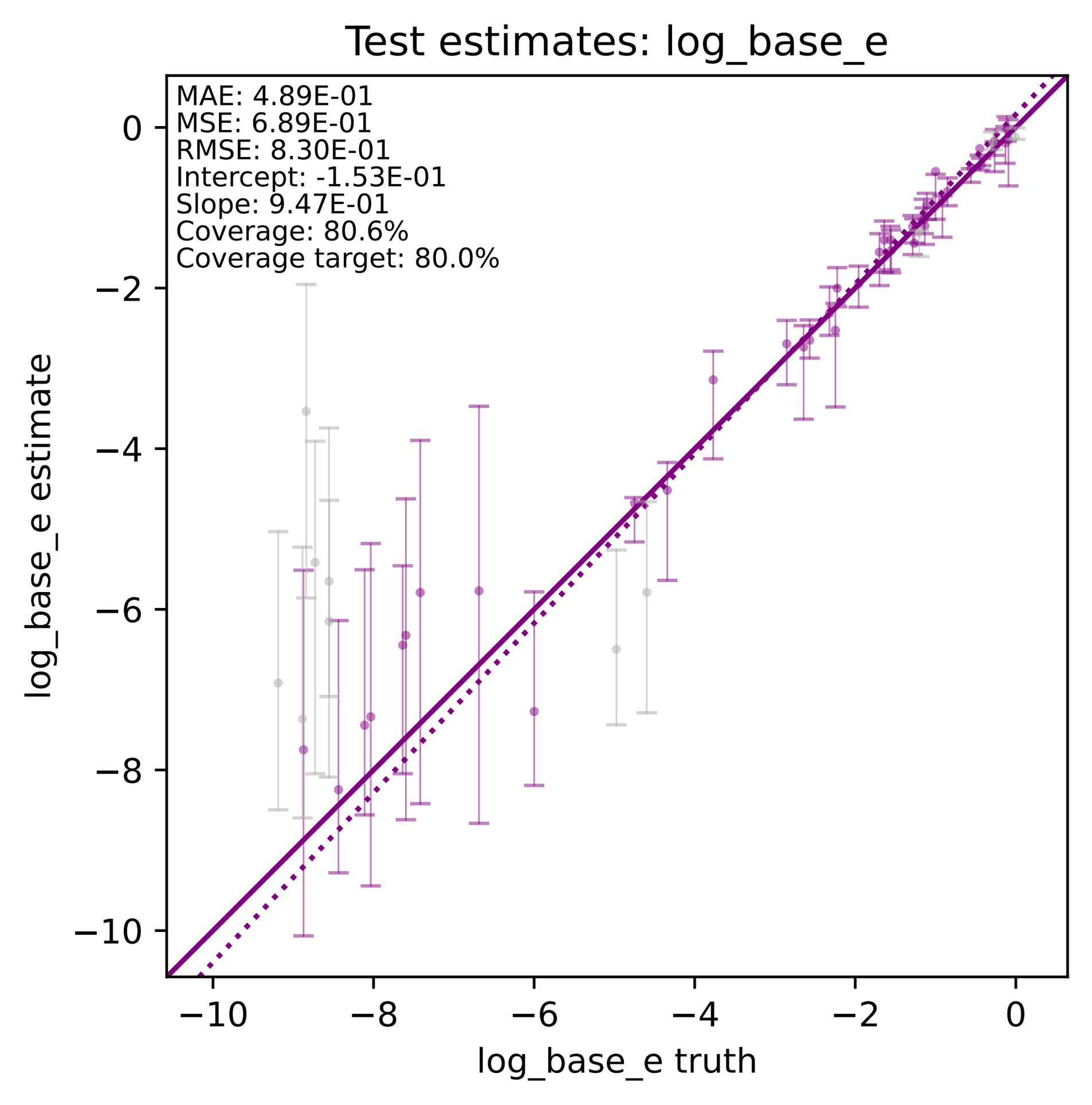

Train estimates: log_base_e
MAE: 4.60E-01
MSE: 5.96E-01
RMSE: 7.72E-01
Intercept: -1.13E-01
Slope: 9.52E-01
Coverage: 81.8%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

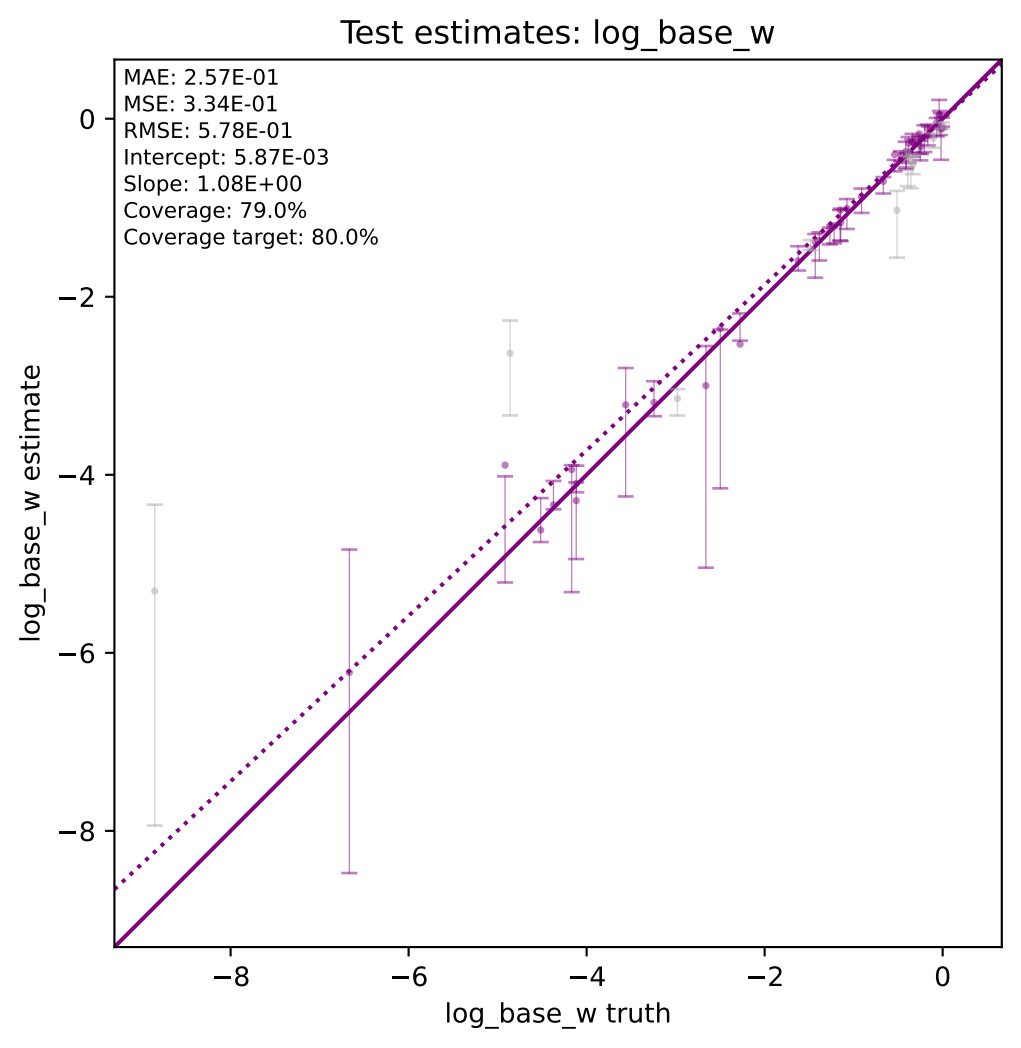

Train estimates: log_base_w
MAE: 2.33E-01
MSE: 2.79E-01
RMSE: 5.28E-01
Intercept: 2.89E-02
Slope: 1.09E+00
Coverage: 82.9%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

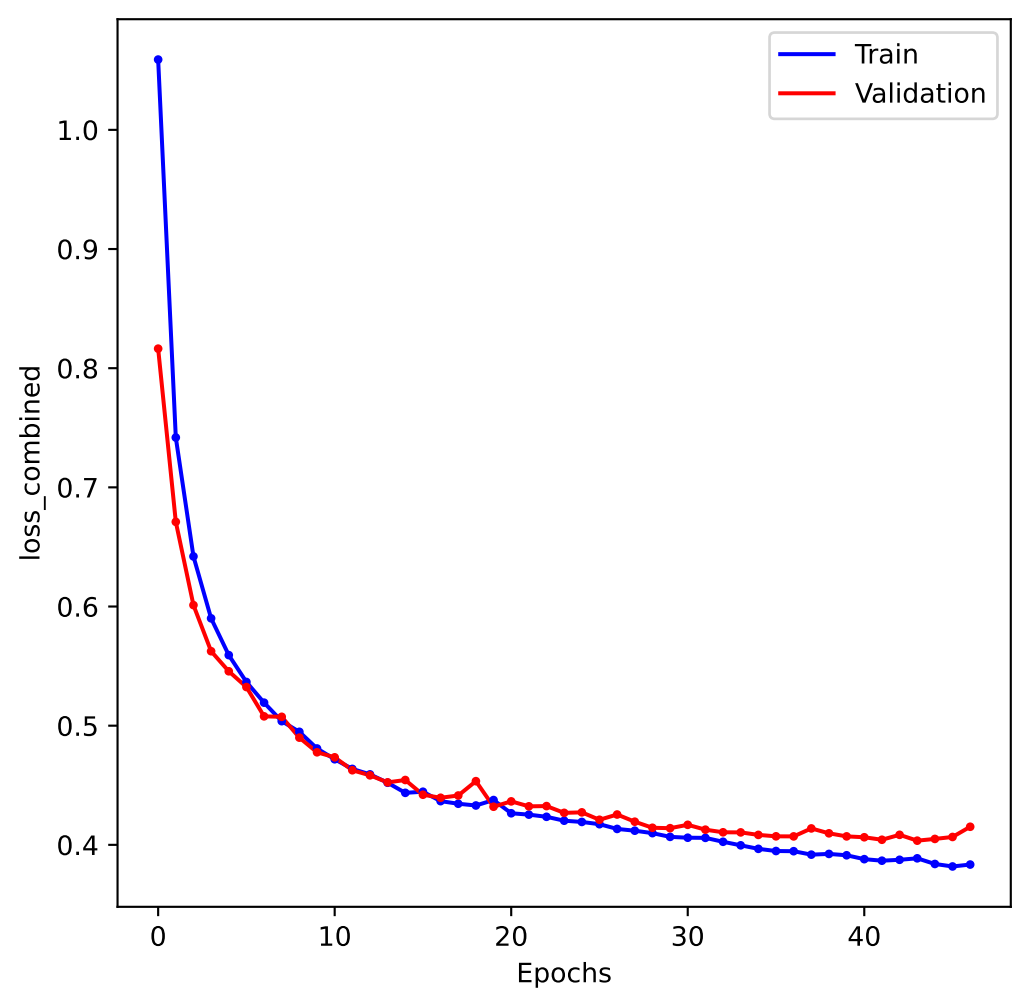

Test estimates: exp_scalar_d_j
MAE: 1.39E-01
MSE: 3.20E-02
RMSE: 1.79E-01
Intercept: 4.73E-02
Slope: 8.70E-01
Coverage: 79.3%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.2
0.4
0.6
0.8
1.0

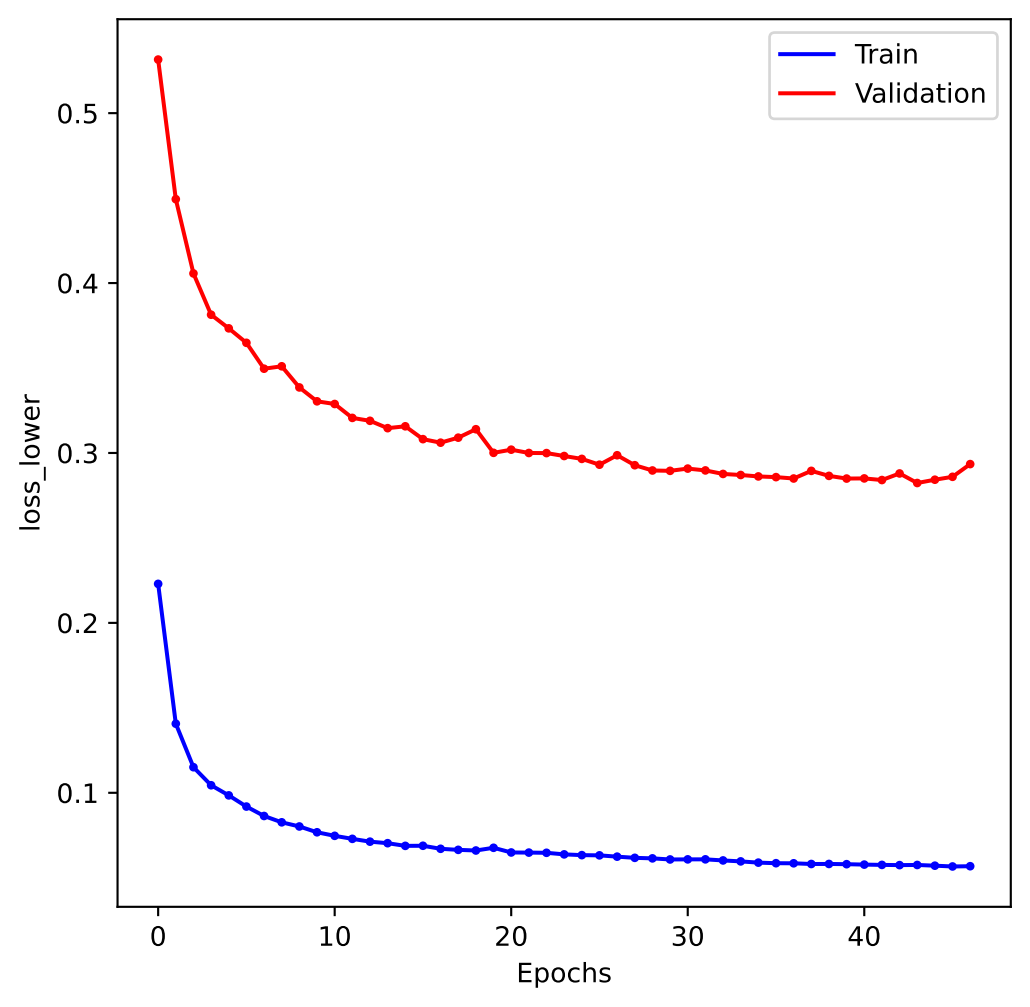

Test estimates: log_base_b
MAE: 9.13E-01
MSE: 1.97E+00
RMSE: 1.40E+00
Intercept: 1.34E-02
Slope: 9.62E-01
Coverage: 81.3%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

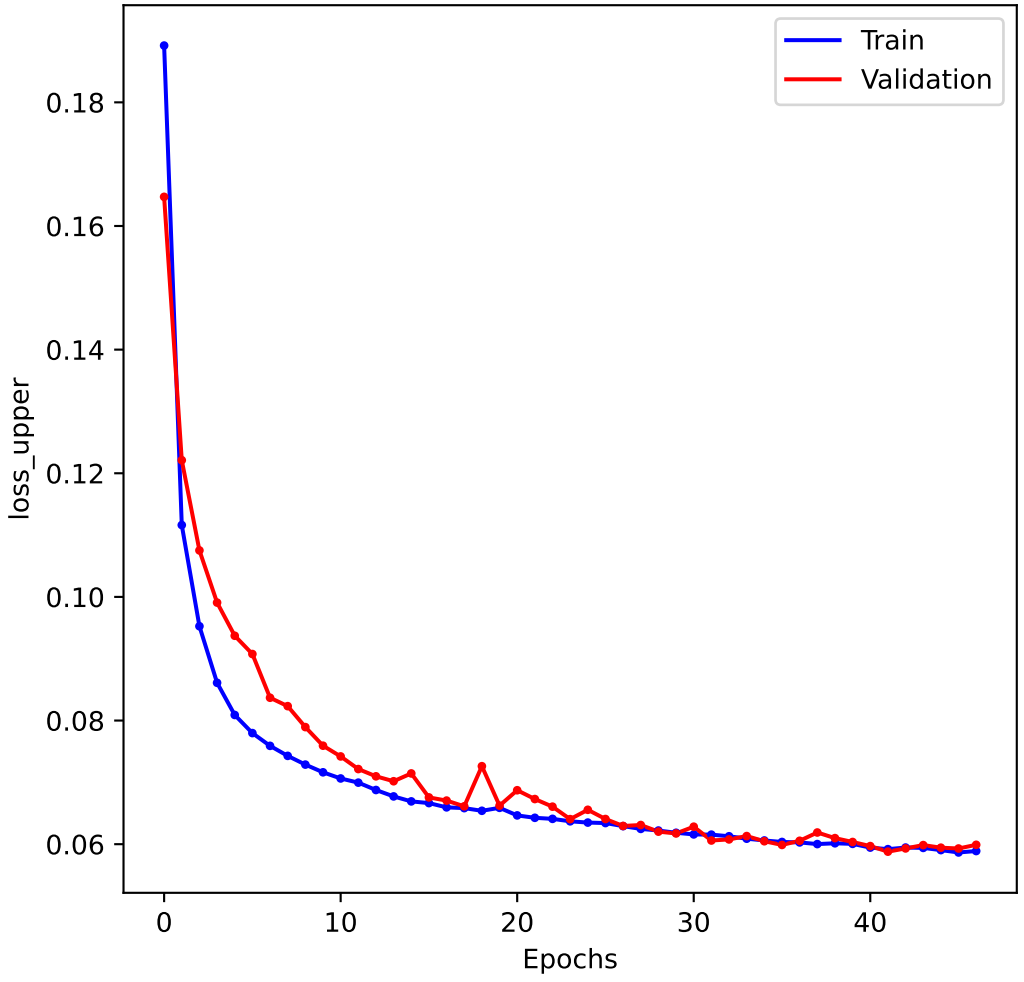

Test estimates: log_base_d
MAE: 4.94E-01
MSE: 4.19E-01
RMSE: 6.47E-01
Intercept: 1.36E-01
Slope: 1.02E+00
Coverage: 79.3%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

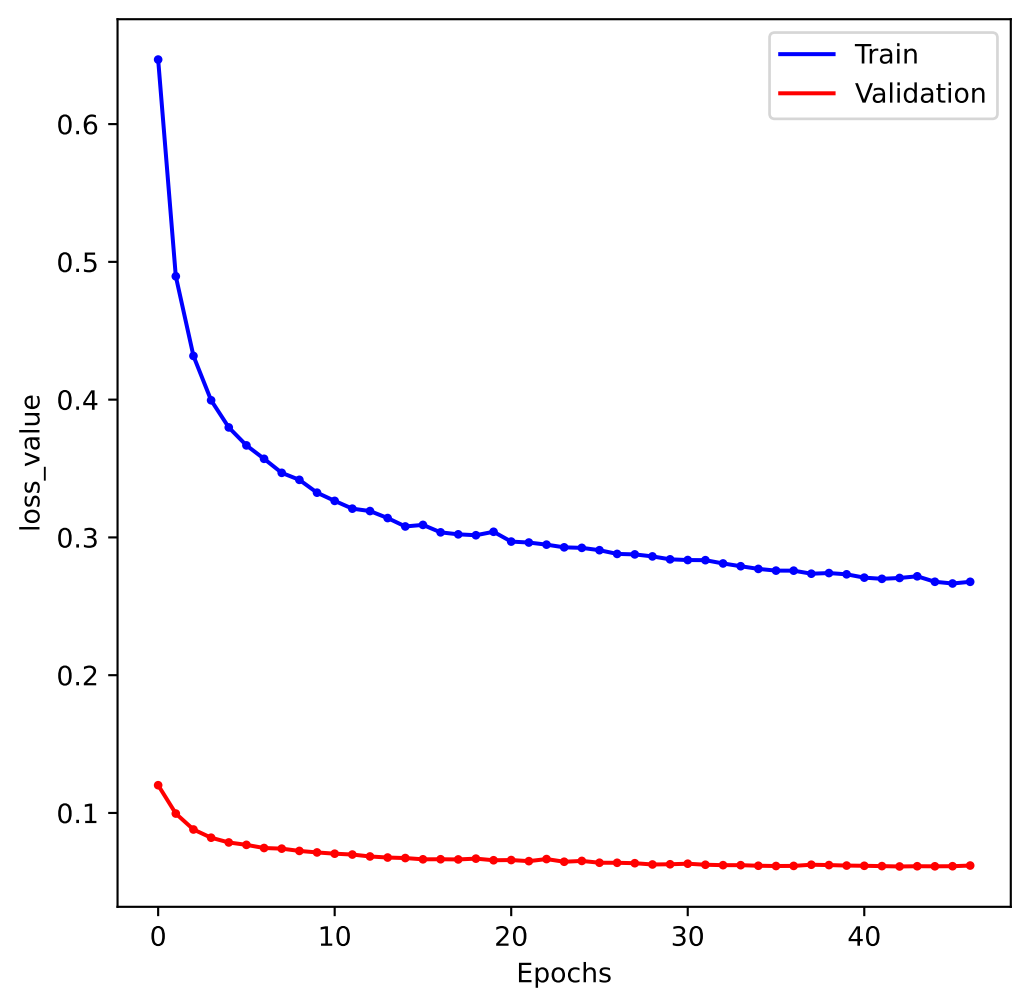

Test estimates: log_base_e
MAE: 4.89E-01
MSE: 6.89E-01
RMSE: 8.30E-01
Intercept: -1.53E-01
Slope: 9.47E-01
Coverage: 80.6%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

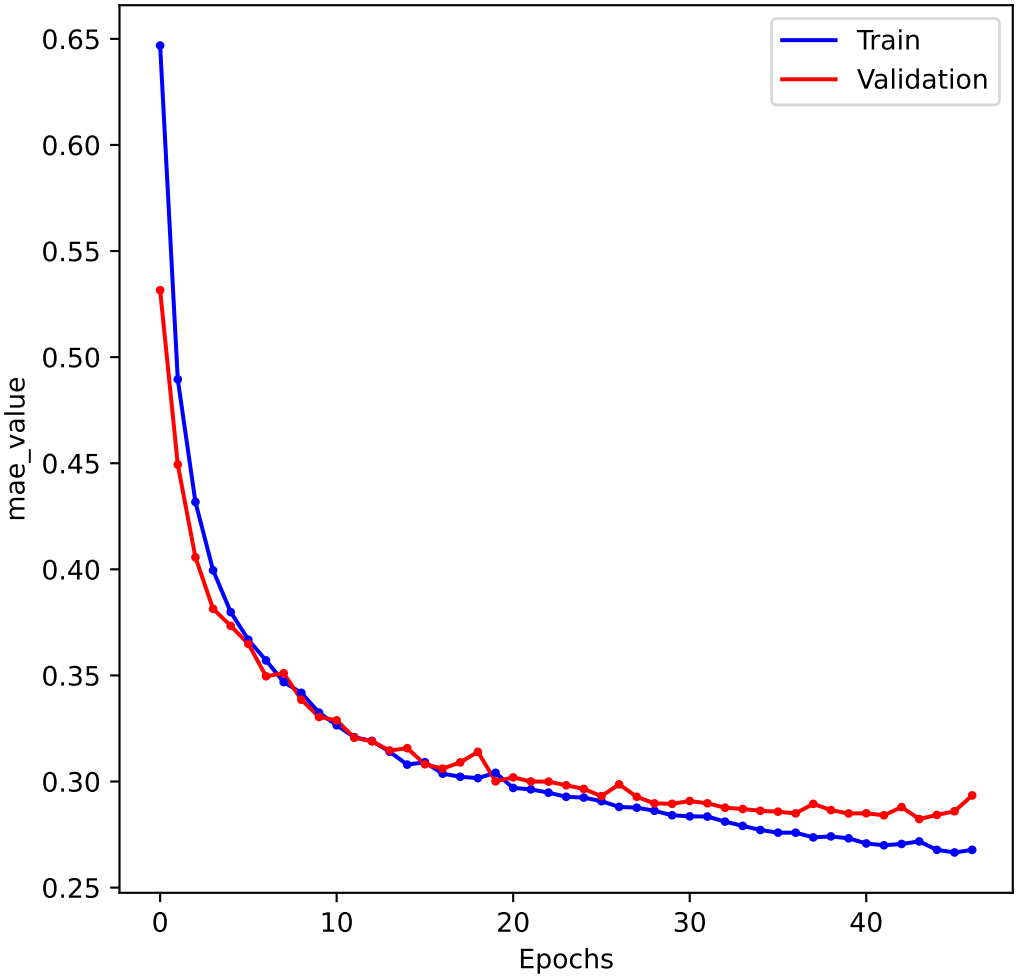

Test estimates: log_base_w
MAE: 2.57E-01
MSE: 3.34E-01
RMSE: 5.78E-01
Intercept: 5.87E-03
Slope: 1.08E+00
Coverage: 79.0%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

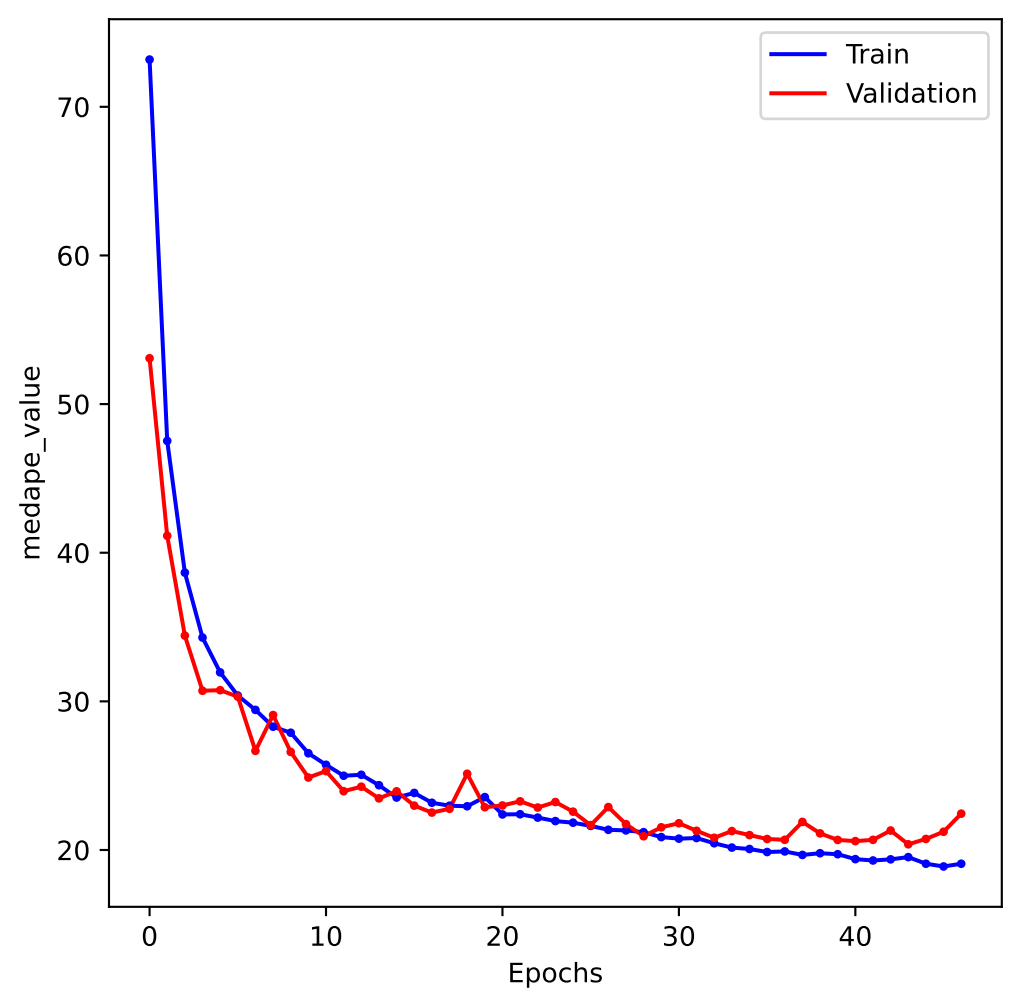

Train
Validation
loss_combined
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0
10
20
30
40
Epochs

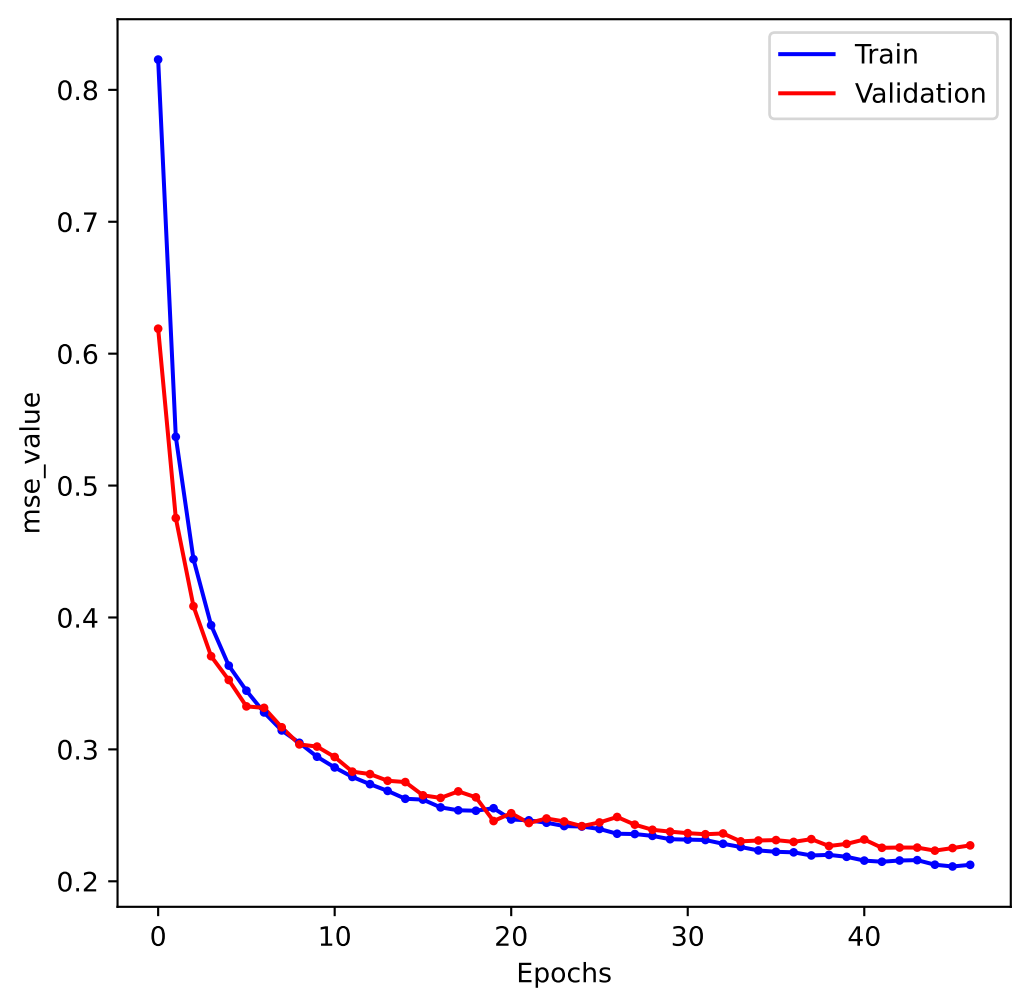

Train
Validation
loss_lower
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
Epochs

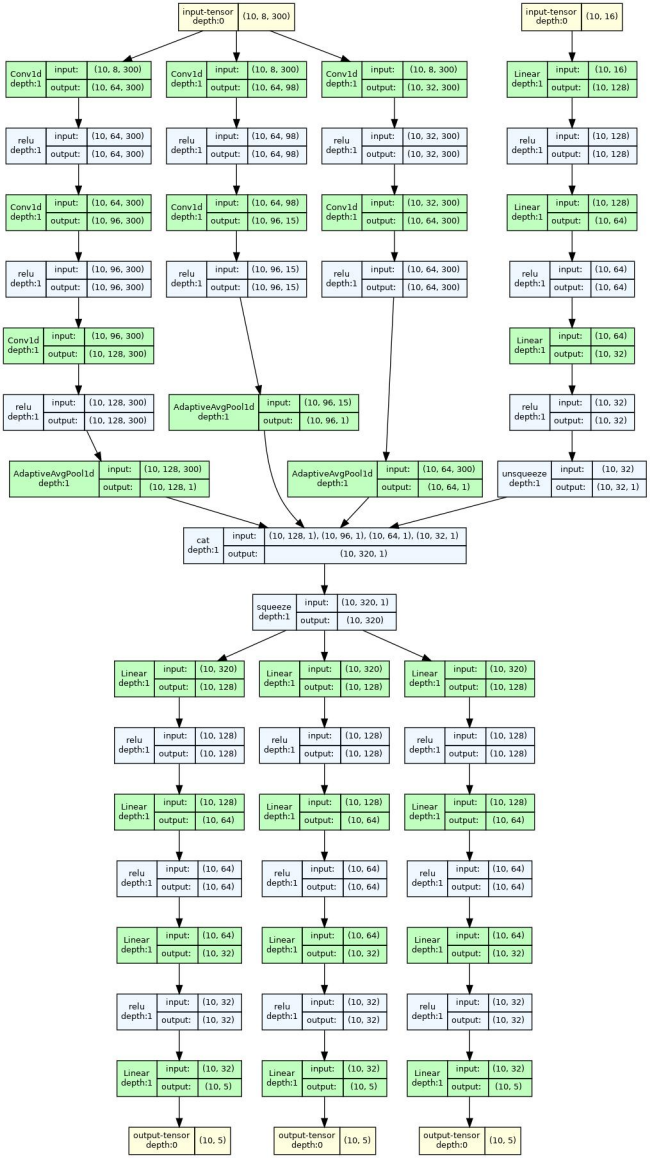

Train
Validation
0.18
0.16
0.14
0.12
0.10
0.08
0.06
loss_upper
0
10
20
30
40
Epochs


Train
Validation
loss_value
0.6
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
Epochs


Train
Validation
mae_value
Epochs
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0
10
20
30
40


Train
Validation
70
60
50
40
30
20
medape_value
0
10
20
30
40
Epochs


Train
Validation
mse_value
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
10
20
30
40
Epochs


input-tensor depth:0
(10, 8, 300)
input-tensor depth:0
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 5)
Linear depth:1
input: (10, 32)
output: (10, 5)
Linear depth:1
input: (10, 32)
output: (10, 5)
output-tensor depth:0
(10, 5)
output-tensor depth:0
(10, 5)
output-tensor depth:0
(10, 5)